

ENGINEERING GRAPHICS

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5ME02	ESC	L	T	P	C	CIA	SEE	Total
		1	-	4	3	30	70	100

COURSE OBJECTIVES

The course should enable the students to:

1. Create awareness and emphasize the need for Engineering Drawing in various branches of engineering.
2. Enable the student with various concepts of dimensioning, conventions and standards related to engineering drawings.
3. Follow the basic drawing standards and conventions.
4. Develop skills in three-dimensional visualization of engineering component.

COURSE OUTCOMES

At the end of the course the student should be able to:

1. Sketch the various curves used in engineering and their applications
2. Apply the knowledge of quadrant system and say to which quadrant and angle of project the object belongs.
3. Evaluate the given object position and draw the projections of objects
4. Analyze the given sectioned objects like in sheet metal applications.
5. Develop the new drawings for the industry requirements

UNIT - I	INTRODUCTION TO ENGINEERING DRAWING	CLASSES: 07
Introduction to Engineering Drawing: Principles of Engineering Graphics and their significance, usage of drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloid, Hypocycloid and Involute.		
UNIT - II	DRAWING OF PROJECTIONS OR VIEWS: ORTHOGRAPHIC PROJECTION IN FIRST ANGLE PROJECTION ONLY	CLASSES: 10
Principles of orthographic projections – conventions – first and third angle projections. Projections of points-Projection of lines inclined to both the planes. PROJECTIONS OF PLANES: Projections of regular planes, inclined to both planes.		
UNIT - III	PROJECTION OF REGULAR SOLIDS	CLASSES: 08
PROJECTION OF SOLIDS-Solids inclined to one plane and both planes (Auxiliary plane method) Sections or Sectional views of Right Regular Solids – Prism, Cylinder, Pyramid, Cone		
UNIT - IV	DEVELOPMENT OF SURFACES/SOLIDS	CLASSES: 04
DEVELOPMENT OF SURFACE/SOLIDS: Theory of development, development of lateral surface along with base		
UNIT - V	ISOMETRIC DRAWINGS	CLASSES: 05
Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections; Drawing Isometric circles, Dimensioning, Isometric Objects; Conversion of Isometric view to Orthographic views and Orthographic to isometric views, Missing views.		
TEXT BOOKS		

1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House
2. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education
3. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication
4. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers .

REFERENCE BOOKS

1. Johle (2009), Engineering Drawing, Tata Mc Graw Hill, New Delhi, India.

WEB REFERENCES

1. nptel.ac.in/courses/112103019/
2. web.iitd.ac.in/~achawla/public_html/201/lectures/sp46.pdf

E-TEXT BOOKS

1. https://www.researchgate.net/publication/305754529_A_Textbook_of_Engineering_Drawing_A_Textbook
2. https://www.researchgate.net/publication/305754529_A_Textbook_of_Engineering_Drawing

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc20_me79/preview